



Web Development Checklist

Comprehensive Quality Assurance Guide

Alpha Solutions Services LLC

Web Development & Digital Solutions

<https://alphasolutions.online/>

Introduction

This comprehensive web development checklist ensures that all critical aspects of your website are thoroughly reviewed before launch. Use this document to maintain high quality standards and deliver exceptional digital experiences to your clients.

How to Use This Checklist:

- Review each section systematically
- Check off items as they are completed and verified
- Document any issues or notes in your project management system

- Ensure all critical items are addressed before deployment

1. Project Planning & Requirements

- Requirements documentation completed and approved
Define all functional and non-functional requirements
- Sitemap and information architecture finalized
Organize content structure and navigation flow
- Wireframes and mockups approved by stakeholders
Visual blueprint of the website layout
- Technical stack and dependencies documented
List all frameworks, libraries, and tools
- Project timeline and milestones established
Set realistic deadlines for each phase
- Budget and resource allocation confirmed
Ensure adequate resources for project completion
- Content strategy and copywriting plan defined
Plan for all text, images, and multimedia
- SEO strategy and keyword research completed
Optimize for search engine visibility

2. Design & User Experience

- Design mockups for all key pages created
Desktop, tablet, and mobile views
- Brand guidelines and style guide followed
Consistent colors, fonts, and imagery
- UI components and design system established
Reusable components for consistency
- Responsive design for all device sizes verified
Test on various screen resolutions
- Accessibility (WCAG 2.1) standards implemented
Ensure usability for all users
- Typography and font loading optimized
Readable text with proper hierarchy

■ Color contrast ratios meet accessibility standards

Minimum 4.5:1 for normal text

■ User flows and interaction patterns tested

Intuitive navigation and user journey

■ Loading animations and micro-interactions added

Enhance user engagement

■ 404 and error page designs completed

Helpful and on-brand error pages

3. Frontend Development

■ HTML semantic markup properly implemented

Use appropriate HTML5 elements

■ CSS organized and maintainable (BEM, SMACSS)

Scalable and modular stylesheets

■ JavaScript code follows best practices

Clean, efficient, and well-documented

■ Cross-browser compatibility tested

Chrome, Firefox, Safari, Edge

■ Mobile responsiveness verified on real devices

Test on actual smartphones and tablets

■ Forms include proper validation and error messages

Client-side and server-side validation

■ Images optimized for web (WebP, lazy loading)

Reduce file sizes without quality loss

■ Icons and graphics implemented (SVG preferred)

Scalable vector graphics for clarity

■ CSS and JavaScript files minified and concatenated

Reduce HTTP requests and file sizes

■ Critical CSS inlined for above-the-fold content

Improve initial page render time ■ Favicon

and app icons created for all platforms *iOS,*

Android, and browser tabs

■ Print stylesheet created if needed

Optimize for printing

4. Backend Development

- Database schema designed and optimized
Proper indexing and relationships
- API endpoints documented and tested
Clear documentation for all APIs
- Authentication and authorization implemented
Secure user login and permissions
- Input validation and sanitization on server side
Prevent SQL injection and XSS
- Error handling and logging configured
Track and debug issues effectively
- Rate limiting and throttling implemented
Prevent abuse and DDoS attacks
- Email functionality tested (contact forms, etc.)
Verify email delivery and formatting
- File upload functionality secured
Validate file types and sizes
- Database backups automated and tested
Regular backups with restore verification
- Environment variables properly configured
Separate dev, staging, production configs
- Third-party API integrations tested
Payment gateways, analytics, etc.
- Session management and cookie handling secure
HTTPOnly, Secure, SameSite flags

5. Content Management

- All content uploaded and proofread
No typos or grammatical errors
- Metadata (title tags, descriptions) optimized
Unique and compelling for each page
- Alt text for all images added
Descriptive text for accessibility and SEO

- Internal linking structure
 - established *Help users and search engines navigate*
- Blog or news section configured if applicable
 - CMS setup and templates*
- Media library organized with proper naming
 - Easy to find and manage assets*
- Copyright notices and legal disclaimers added
 - Protect intellectual property*
- Privacy policy and terms of service published
 - Legal compliance*
- Contact information accurate and up-to-date
 - Phone, email, address, social media*
- Social media links functional and correct
 - Link to active social profiles*

6. Performance Optimization

- Page load time under 3 seconds (target)
 - Measure with Google PageSpeed Insights*
- Images compressed and properly sized
 - Use modern formats like WebP*
- CDN configured for static assets
 - Faster delivery worldwide*
- Gzip or Brotli compression enabled
 - Reduce transfer sizes*
- Browser caching configured properly
 - Leverage browser cache*
- Database queries optimized
 - Use indexing and avoid N+1 queries*
- Unnecessary plugins and scripts removed
 - Reduce bloat and conflicts*
- Async and defer attributes used for scripts
 - Non-blocking JavaScript loading*
- Web fonts optimized (subset, preload)
 - Reduce font file sizes*

- Third-party scripts evaluated for performance impact

Remove or defer heavy scripts

- Lighthouse performance score 90+ (target)

Google's performance benchmark

- Core Web Vitals pass thresholds

LCP, FID, CLS metrics

7. Security

- SSL/TLS certificate installed and configured

HTTPS enabled sitewide

- Security headers implemented (CSP, X-Frame-Options)

Protect against common attacks

- SQL injection protection verified

Parameterized queries

- Cross-site scripting (XSS) prevention tested

Escape output, sanitize input

- Cross-site request forgery (CSRF) tokens used

Protect form submissions

- Sensitive data encrypted (passwords, etc.)

Use bcrypt or Argon2

- File upload security measures in place

Validate types, scan for malware

- Admin area protected (strong passwords, 2FA)

Secure administrative access

- Regular security updates scheduled

Keep software up-to-date

- Dependencies scanned for vulnerabilities

Use npm audit, Snyk, etc.

- Backup and disaster recovery plan documented

How to restore from backup

- DDoS protection configured if needed

Cloudflare or similar service

8. SEO & Marketing

- XML sitemap generated and submitted

Submit to Google Search Console

- Robots.txt file configured correctly
Allow/disallow appropriate paths
- Canonical URLs implemented
Prevent duplicate content issues
- Open Graph tags for social sharing added
Control appearance on social media
- Twitter Card meta tags implemented
Optimize for Twitter sharing
- Schema.org structured data added
Help search engines understand content
- Google Analytics or alternative tracking installed
Monitor traffic and user behavior
- Google Search Console configured
Monitor search performance
- Goal tracking and conversion funnels set up
Measure success metrics
- 404 errors monitored and redirected
Fix broken links
- Redirect strategy for old URLs implemented
301 redirects for moved content
- Local SEO optimized (if applicable)
Google My Business, local citations

9. Accessibility (WCAG 2.1)

- Keyboard navigation fully functional
All interactive elements accessible
- Screen reader compatibility tested
Use NVDA or JAWS
- ARIA labels and roles properly used
Enhance semantic meaning
- Skip navigation link provided
Allow users to bypass repetitive content
- Focus indicators visible and clear
Show which element has focus

- Videos include captions and transcripts
Accessible multimedia content
- Forms have associated labels
Proper label-input relationships
- Error messages are clear and helpful
Guide users to fix issues
- Text can be resized up to 200% without breaking layout
Browser zoom compatibility
- No content flashes more than 3 times per second
Prevent seizures
- Accessibility statement or policy published
Communicate commitment to accessibility

10. Testing & Quality Assurance

- Cross-browser testing completed
Latest versions of major browsers
- Mobile device testing on multiple
devices *iOS, Android, various screen sizes*
- Functional testing of all features
Forms, search, navigation, etc.
- User acceptance testing (UAT) conducted
Stakeholder approval
- Broken link checker run
No 404 errors internally
- Spell check and grammar review performed
Professional content quality
- Load testing under expected traffic
Ensure site handles traffic spikes
- Usability testing with real users
Gather feedback on user experience
- A/B testing plan for key pages (if applicable)
Optimize conversion rates

- Regression testing after bug fixes

Ensure fixes don't break other features

- Performance testing on slow connections

Test on 3G, 4G networks

- Checkout process tested end-to-end (e-commerce)

Complete purchase flow

11. Legal & Compliance

- GDPR compliance implemented (if applicable)

Cookie consent, data protection

- CCPA compliance implemented (if California users)

California privacy rights

- Cookie consent banner configured

Inform users about cookie usage

- Privacy policy clearly accessible

Link in footer

- Terms and conditions published

User agreement

- Accessibility statement published

Commitment to accessibility

- Copyright notices on all pages

Protect intellectual property

- Third-party licenses documented

Comply with open source licenses

- Data processing agreements with vendors

GDPR Article 28 compliance

- User data deletion process established

Right to be forgotten

- Age verification for restricted content (if needed)

Comply with age restrictions

12. Launch Preparation

- Staging environment reviewed by stakeholders

Final approval before launch

- DNS settings configured and tested

Point domain to correct server

■ Email accounts set up and tested

Professional email addresses

■ Monitoring and alerting configured

Uptime monitoring, error tracking

■ Deployment process documented

Step-by-step launch procedure

■ Rollback plan

prepared How to

revert if issues arise

■ Post-launch support plan established

Who handles issues after launch

■ Client training completed (if applicable)

CMS training, content updates

■ Launch announcement prepared

Social media, email, press release

■ Search engine re-crawl requested

Submit updated sitemap

■ Old site redirects tested (if redesign)

301 redirects from old URLs

■ Temporary maintenance page ready (if needed)

For planned downtime

13. Post-Launch Monitoring

■ Monitor server performance and uptime

First 48 hours critical

■ Check analytics for traffic patterns

Verify tracking is working

■ Monitor error logs for issues

Fix critical bugs immediately

■ Review user feedback and support tickets

Address user concerns

■ Check search engine indexing status

Verify pages are being indexed

- Monitor social media mentions
Respond to comments and questions
 - Review and optimize slow-loading pages
Continuous performance improvement
 - Schedule regular content updates
Keep site fresh and relevant
 - Plan for ongoing maintenance and updates
Security patches, feature additions
 - Gather user feedback for future improvements
Surveys, heatmaps, session recordings
 - Review and update documentation
Keep technical docs current
 - Schedule periodic security audits
Quarterly or bi-annual reviews
-